

MECHANICS OF MATERIALS AND STRUCTURES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.632
Applied Finite Elements

Every Term

CAD/FEA

535.731
Engineering Materials: Properties
and Selections

Fall/Spring

Materials Science

535.623
Intermediate Vibrations

Fall/Spring

Dynamics/Controls

535.606
Advanced Strength of Materials

Spring

535.607
Mechanics of Solids & Structures:
Theory and Applications I

Fall

Mechanics

MECHANICS OF MATERIALS AND STRUCTURES

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 2 Courses

535.627
Computer-Aided Design

Every Term

CAD

535.706
Mechanics of Solids & Structures:
Theory and Applications II

Spring

535.663
Biosolid Mechanics

Spring/Summer

535.732
Fatigue and Fracture of Materials

Spring

585.631
Introduction to Biomechanics

Spring

Biomechanics

535.748
Stress Waves, Impacts and
Shockwaves

Spring

535.643
Plasticity

Spring

535.612
Intermediate Dynamics

Fall

565.604
Structural Mechanics

Fall

Mechanics

585.631
Introduction to Biomechanics

Spring

Dynamics

535.660
Precision Mechanical Design

Fall/Spring

565.682
Design of Ocean Structures

Summer (Odd years)

675.722
Space Mechanical Systems
Design and Analysis
In Development

Machine & Structure Design

515.602
Thermodynamics and Kinetics of
Materials

Spring

Thermodynamic

525.606
Electronic Materials

Spring

Electronic Materials

535.720
Mechanics of Composite
Materials and Structures

Fall

535.721
Advanced Composite Materials &
Manufacturing Processes

Spring/Summer

535.733
Micromechanics of
Heterogeneous Materials &
Composites

Fall

Composites

535.684
Modern Polymeric Materials

Summer/Fall

515.661
Introduction to Polymer Science

Fall

545.660
Polymer Physics

In Development

Polymer Science

535.618
Fabricatology: Advanced
Materials Processing

Summer/Fall

535.674
Smart Manufacturing and
Automation

Spring

515.658
Design for Additive Manufacturing

Fall

Additive

535.742
Applied Machine Learning for
Mechanical Engineers

Spring/Summer

535.743
Intermediate Applied AI in
Mechanical Engineering

Fall

535.744
AI for Mechanical Materials
Design

Spring

AI/ML

515.601
Structure and Properties of
Materials

Fall

515.617
Nanomaterials

Spring/Summer

515.622
Micro and Nano Structured
Materials & Devices

TBD

525.606
Electronic Materials

In Development

675.702
Materials for Space Systems

Every Term

Materials

535.737
Multiscale Modeling and
Simulation of Mechanical Systems

Summer

Computation